

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Agile Sprint Planning & User Story Workshop
Weightage:	20%	Total Marks:	25
CO1 Statement:	<i>Apply advanced methodologies and agile project management techniques to manage software projects effectively</i>		
Student Name:	K V CHARAN TEJA	Reg. No.:	192424167
Date:	27-0-2026	Duration:	60 minutes

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

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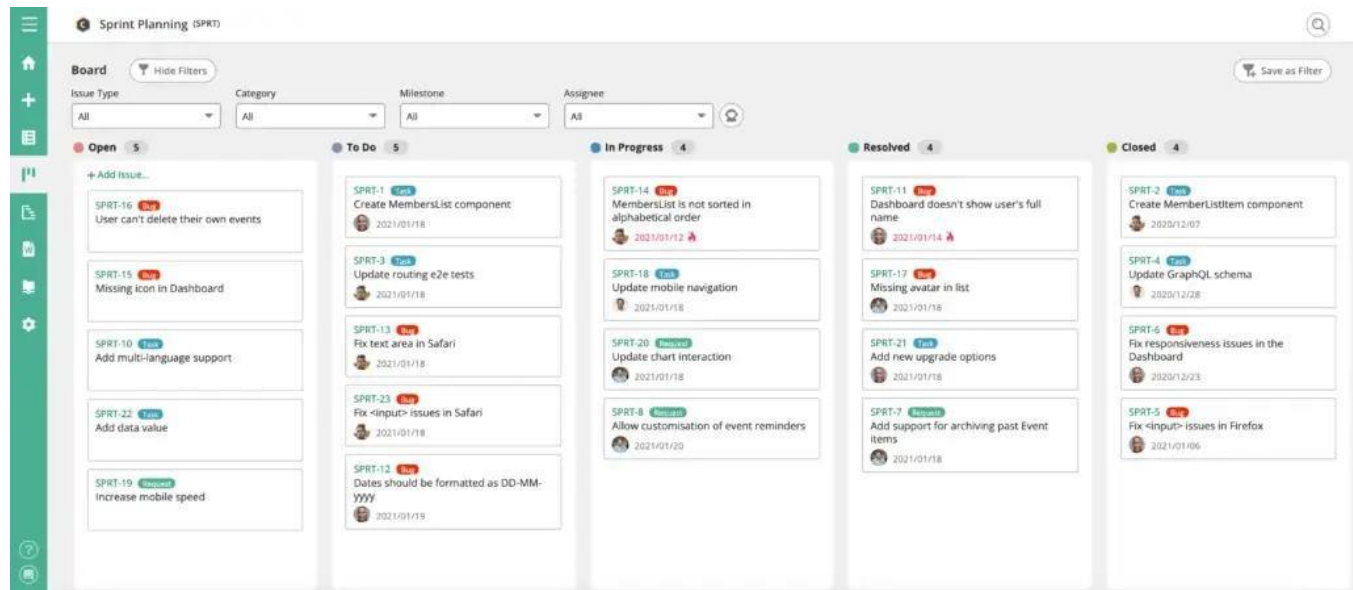
- 1. Identify the Project Vision and Stakeholders**
 - Define the project's vision, objectives, and identify all key stakeholders involved.
- 2. Create Epics and User Stories**
 - Break down the requirements into Epics and formulate well-structured User Stories using the format:
As a <user>, I want <feature>, so that <benefit>.
- 3. Define Acceptance Criteria**
 - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
- 4. Prioritize the Product Backlog**
 - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
- 5. Plan the Sprint Backlog**
 - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
- 6. Estimate Effort Using Story Points**
 - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
- 7. Present the Sprint Goal and Expected Deliverables**
 - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Domain: Banking & Finance

Project: Blockchain-Based Secure Banking Transaction Management System

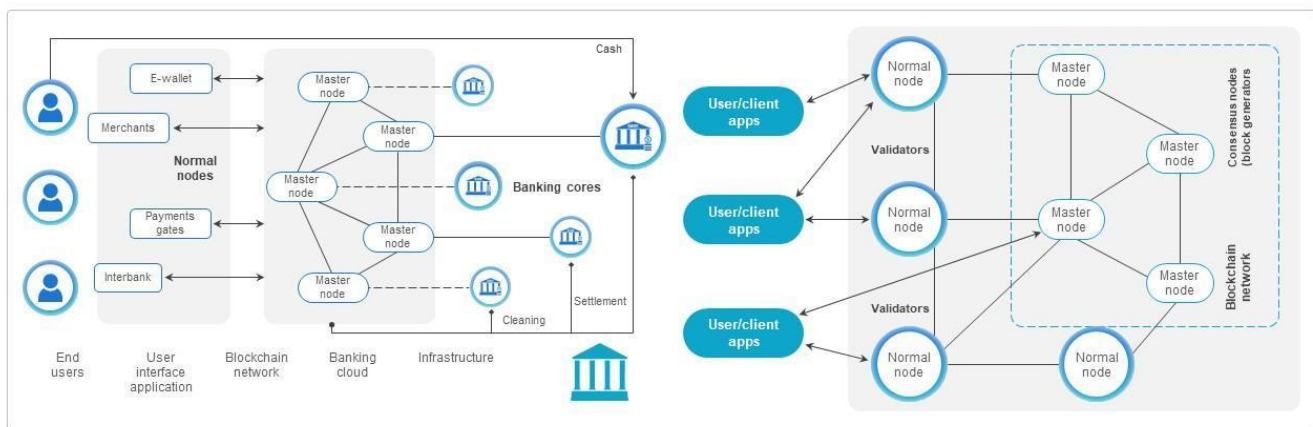
Agile Sprint Planning & User Story Workshop

Blockchain-Based Secure Banking Transaction Management System



Blockchain enabled cloud computing architecture for banks

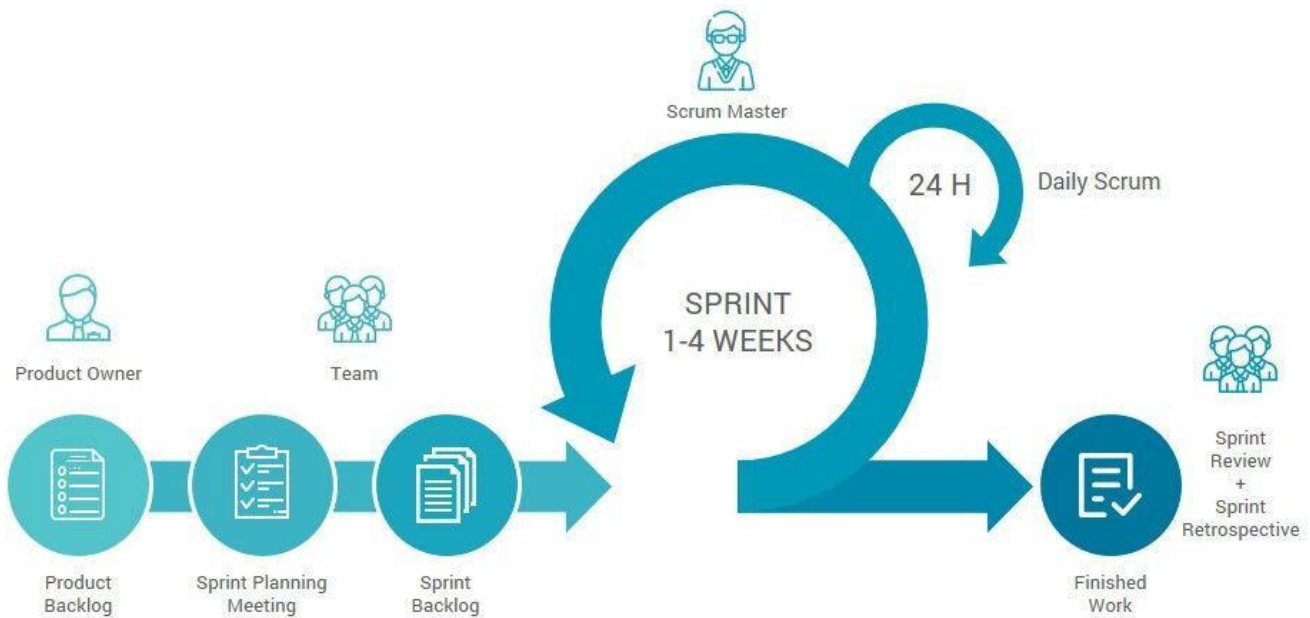
This slide provides Architecture for Blockchain-based Cloud banking ecosystem aiming to enhance operations. Key components involved include end users, user interface applications, blockchain network etc.



System comprises two classes of nodes: master nodes and normal nodes, each with distinct rights and roles

Master nodes have the highest authority, responsible for generating blocks, executing core protocols, and implementing the consensus mechanism.

Normal nodes, including banks, financial institutions, payment service providers, and major merchants, serve as transaction validators



4

1. Project Vision

Vision Statement

The vision of the **Blockchain-Based Secure Banking Transaction Management System** is to develop a secure, transparent, decentralized, and fraud-resistant banking platform using blockchain technology. The system ensures secure financial transactions, immutable transaction records, faster payment processing, and enhanced customer trust while reducing cyber threats and operational costs.

Project Objectives

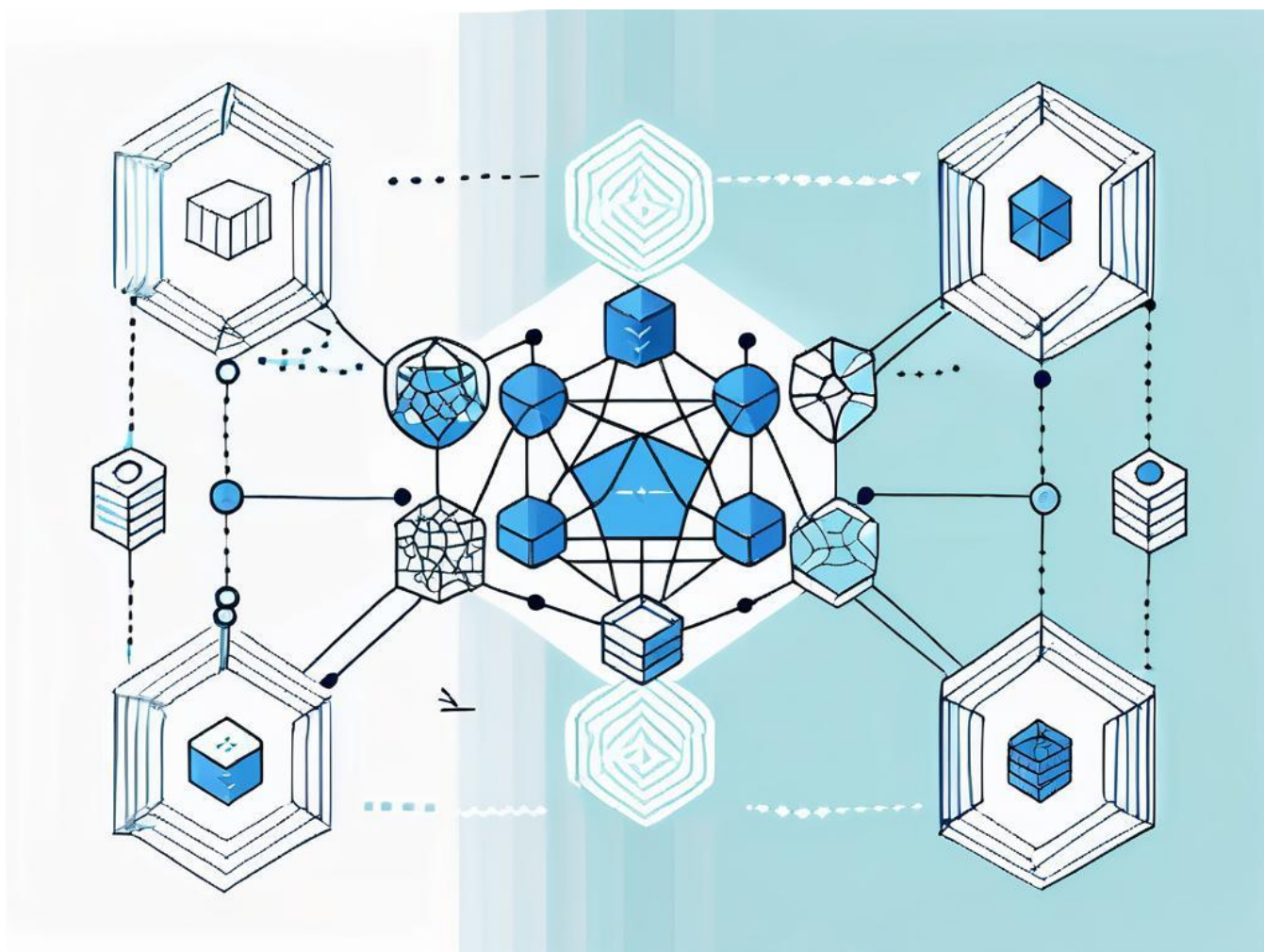
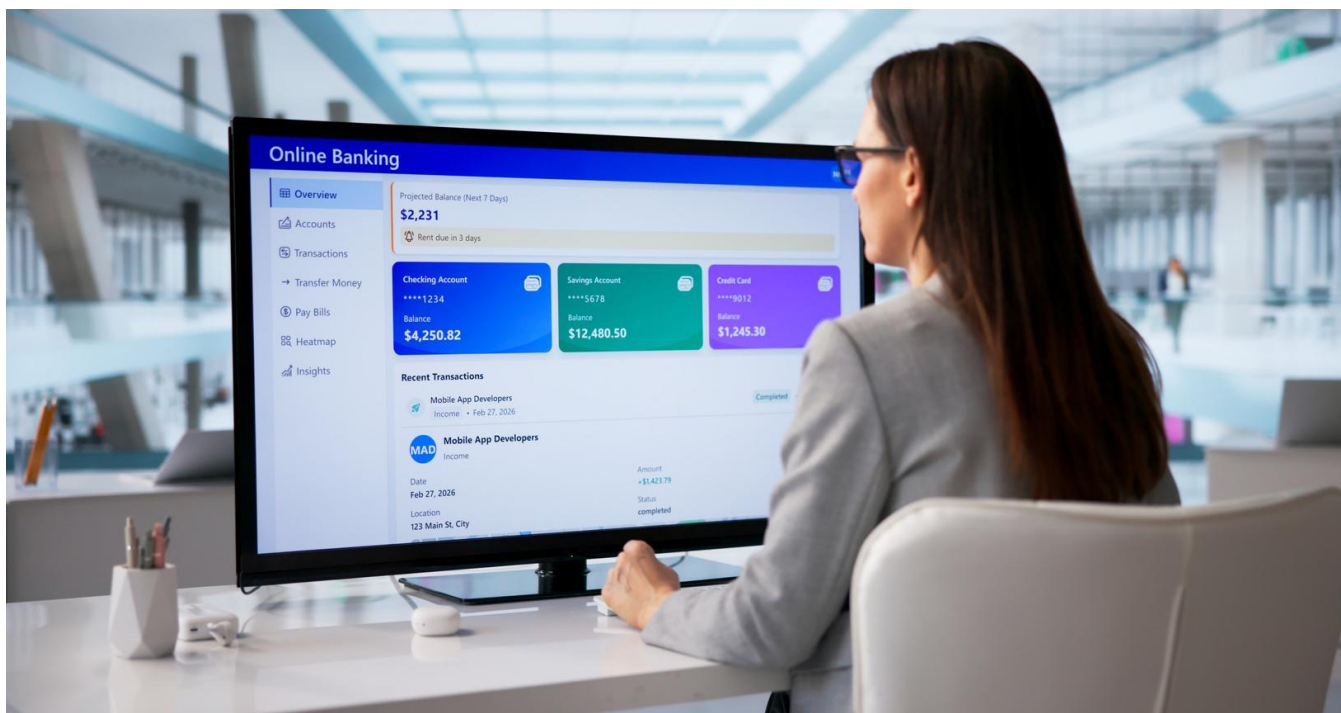
- Develop a decentralized banking system.
 - Secure all banking transactions using blockchain.
 - Prevent fraud and unauthorized access.
 - Enable real-time transaction verification.
 - Protect customer financial information.
 - Improve transparency and accountability.
 - Automate transaction validation using smart contracts.
 - Provide 24×7 secure online banking services.
-

Stakeholders

Stakeholder	Role
Customer	Performs banking transactions
Bank Employee	Verifies customer requests
Bank Manager	Monitors banking activities
System Administrator	Maintains the system
Blockchain Validators	Validate transactions
Cyber Security Team	Protects against attacks
Government Authorities	Regulatory compliance
Financial Institution	Banking operations

Banking Stakeholders





2. Epics & User Stories

Epic 1 – Customer Account Management

User Story 1

As a customer, I want to register an account so that I can access secure banking services.

User Story 2

As a customer, I want to log in using multi-factor authentication so that my account remains secure.

Epic 2 – Banking Transactions

User Story 3

As a customer, I want to transfer money securely so that transactions are completed safely.

User Story 4

As a customer, I want to view my transaction history so that I can monitor all financial activities.

Epic 3 – Blockchain Security

User Story 5

As a blockchain validator, I want to verify transactions so that fraudulent activities are prevented.

User Story 6

As a system administrator, I want every transaction stored in an immutable blockchain ledger so that records cannot be modified.

Epic 4 – Fraud Detection

User Story 7

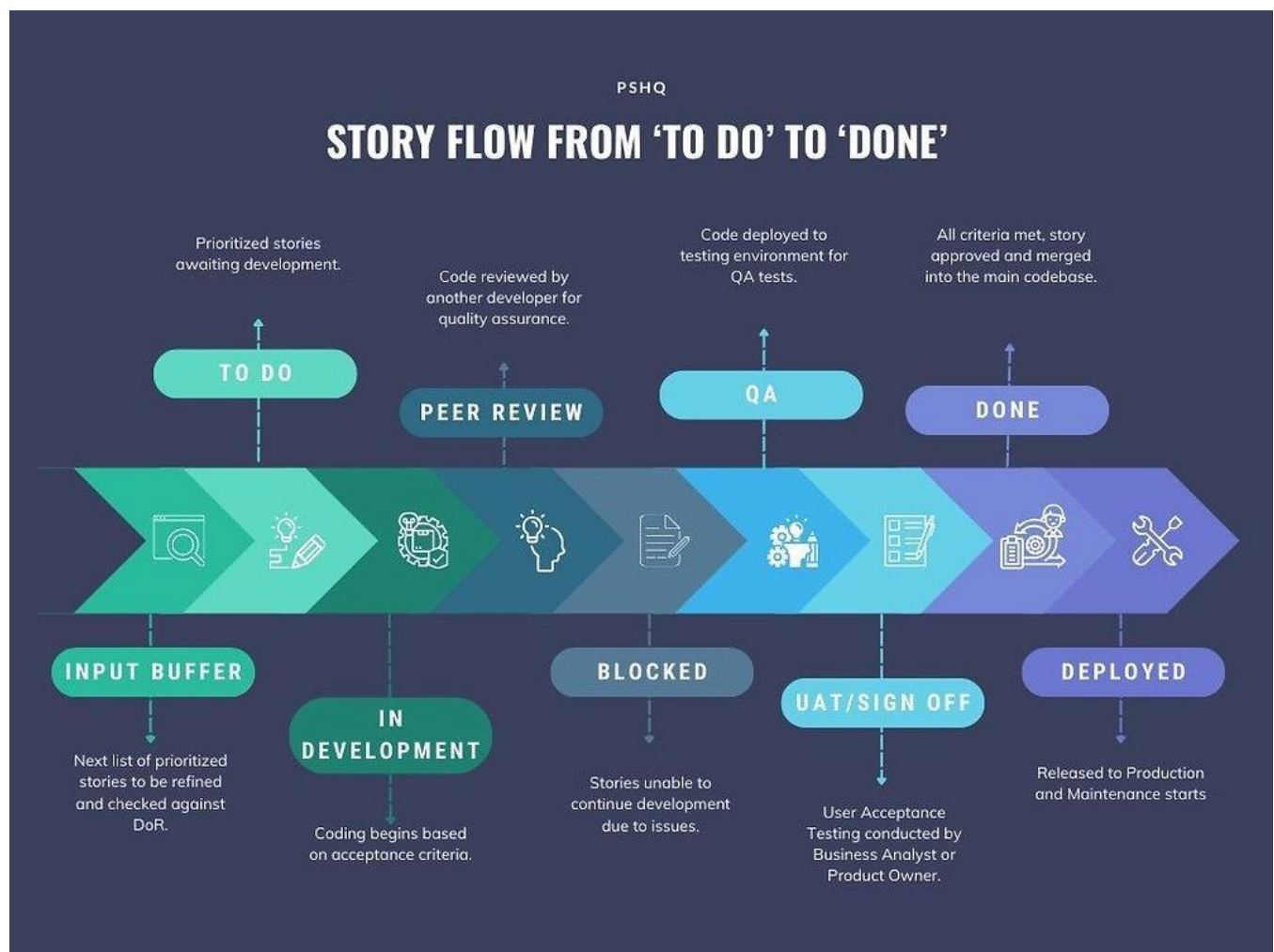
As a bank manager, I want suspicious transactions to be detected automatically so that fraud can be prevented.

Epic 5 – Reporting

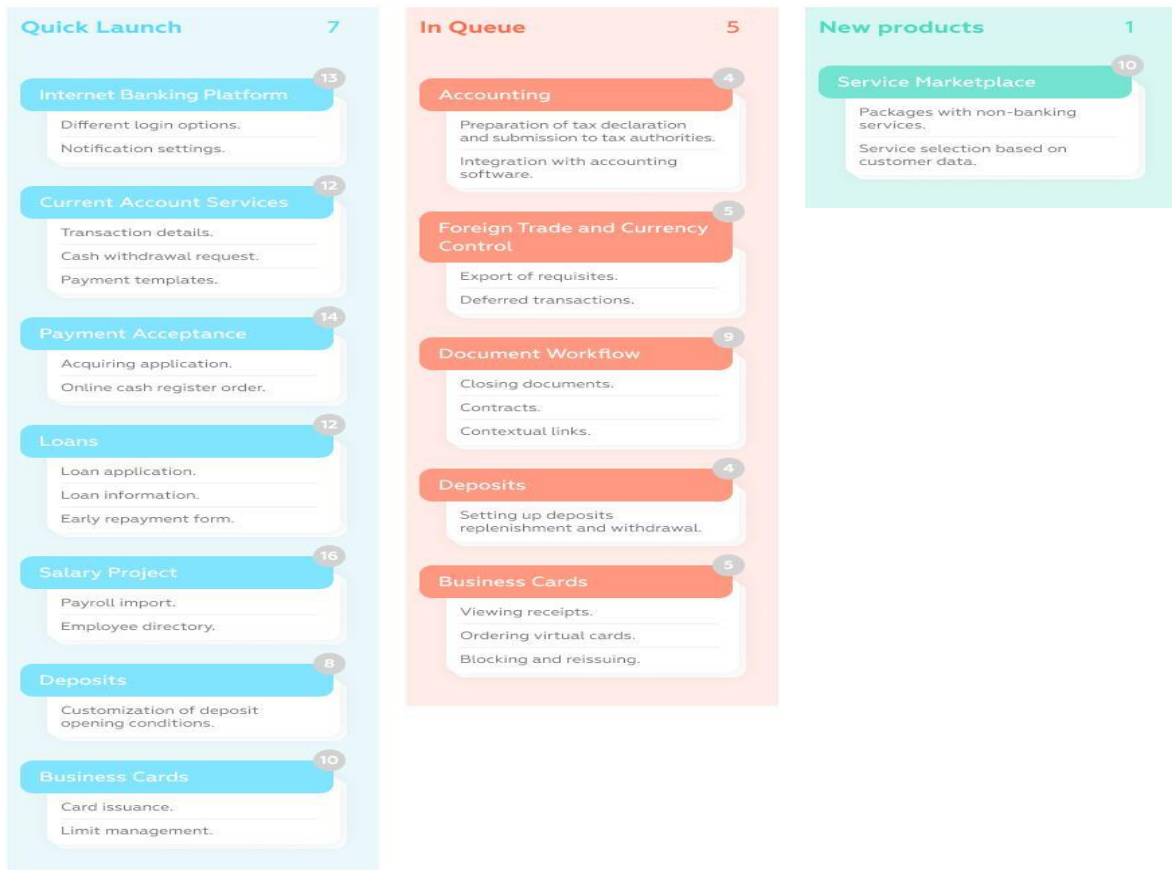
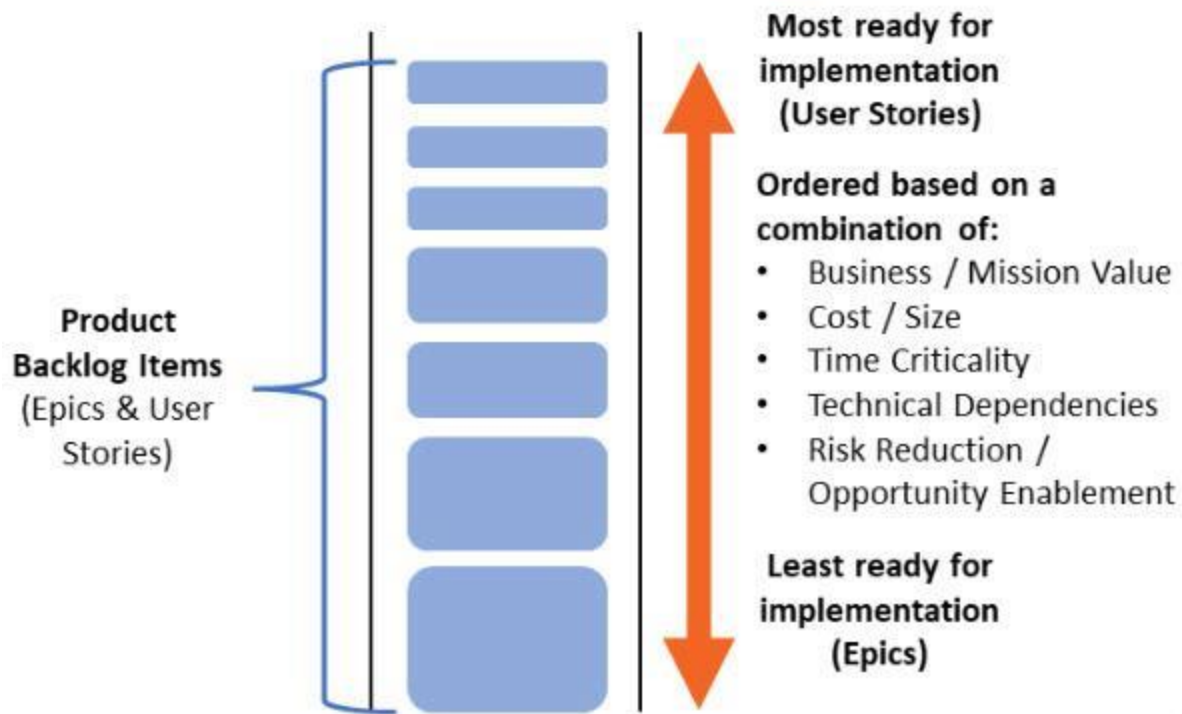
User Story 8

As an administrator, I want to generate audit reports so that regulatory compliance can be maintained.

Agile User Story Workflow



Product Backlog



3. Acceptance Criteria

User Story 1

Given

The customer opens the registration page.

When

The customer enters valid personal details.

Then

The account should be created successfully.

User Story 2

Given

The customer enters login credentials.

When

OTP verification is successful.

Then

The customer should access the dashboard.

User Story 3

Given

The customer has sufficient balance.

When

The customer initiates a transfer.

Then

The blockchain validates the transaction and updates both accounts successfully.

User Story 4

Given

The customer opens transaction history.

When

The request is submitted.

Then

All previous blockchain transactions should be displayed.

User Story 5

Given

A transaction is created.

When

Blockchain validators verify the transaction.

Then

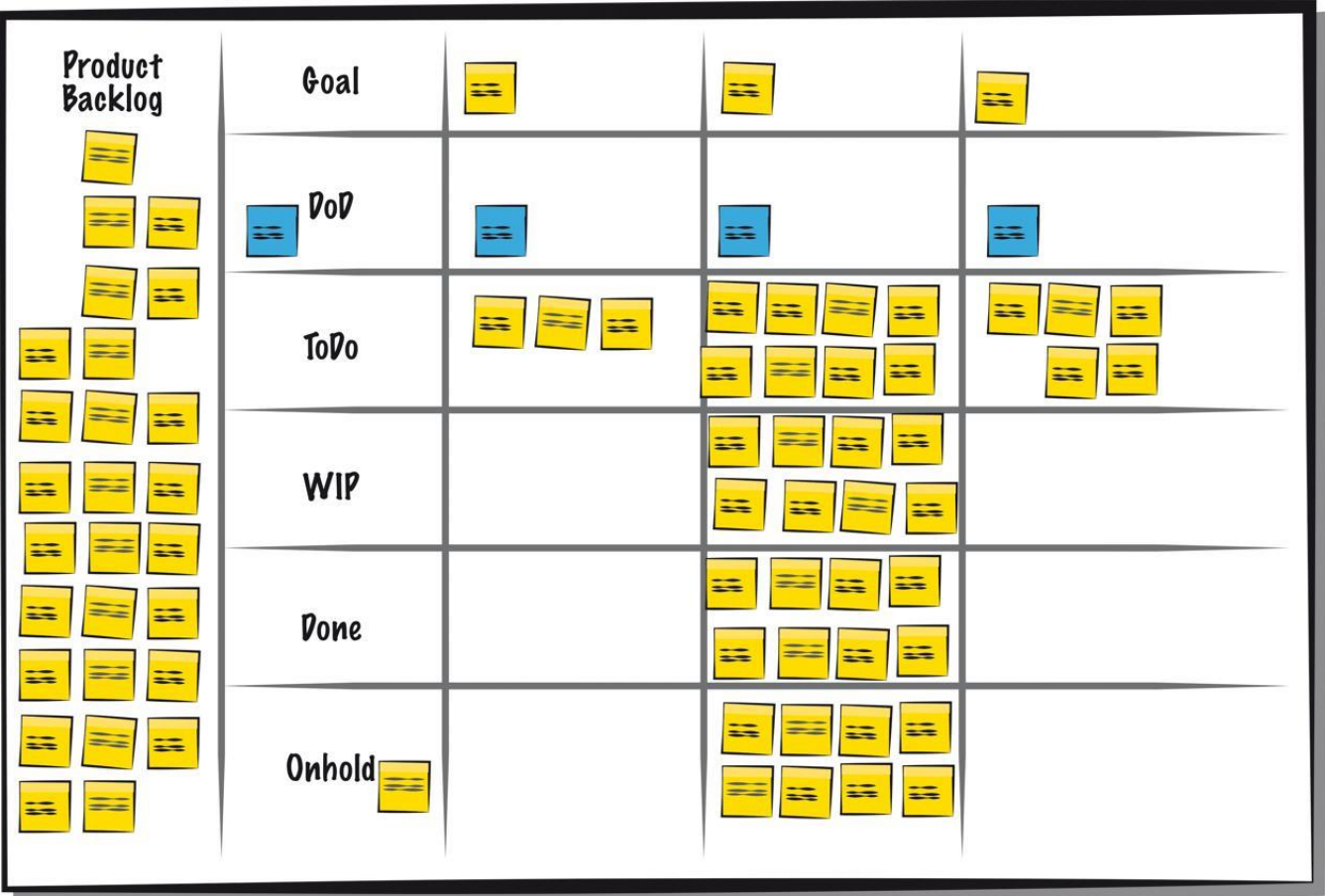
The transaction should be permanently stored in the blockchain ledger.

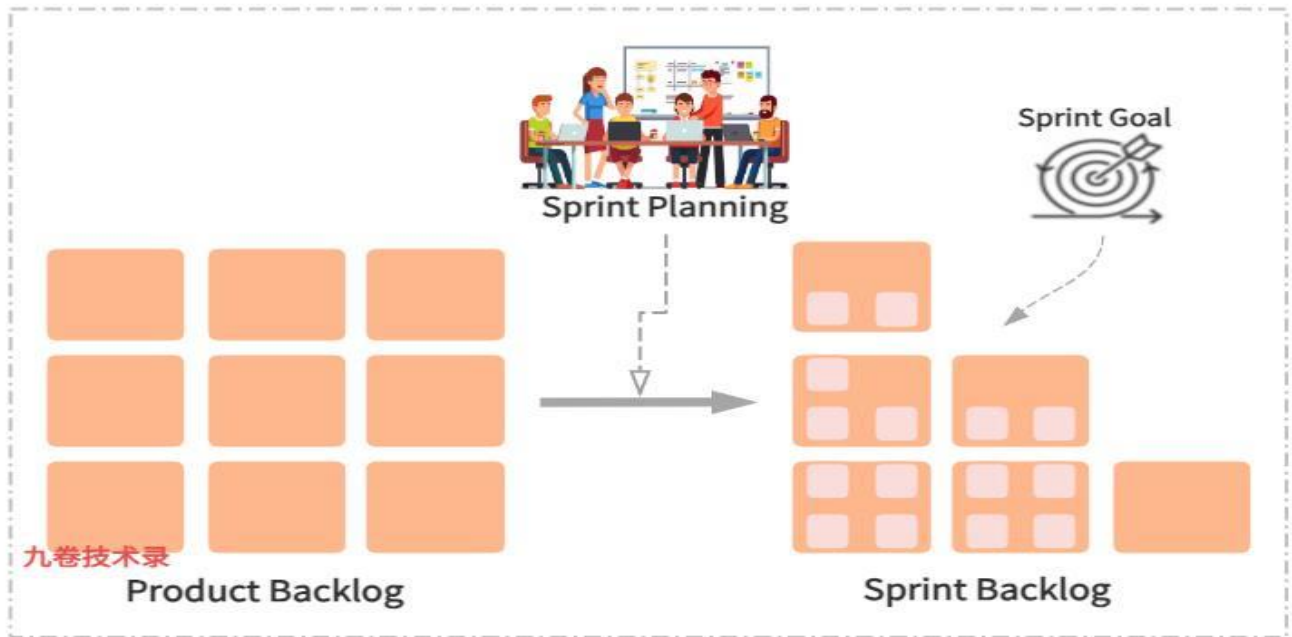
4. Product Backlog

Priority	Product Backlog Item
High	Customer Registration
High	Secure Login
High	Blockchain Validation
High	Fund Transfer
High	Smart Contracts
Medium	Transaction History
Medium	Fraud Detection
Medium	Notifications
Low	Audit Reports

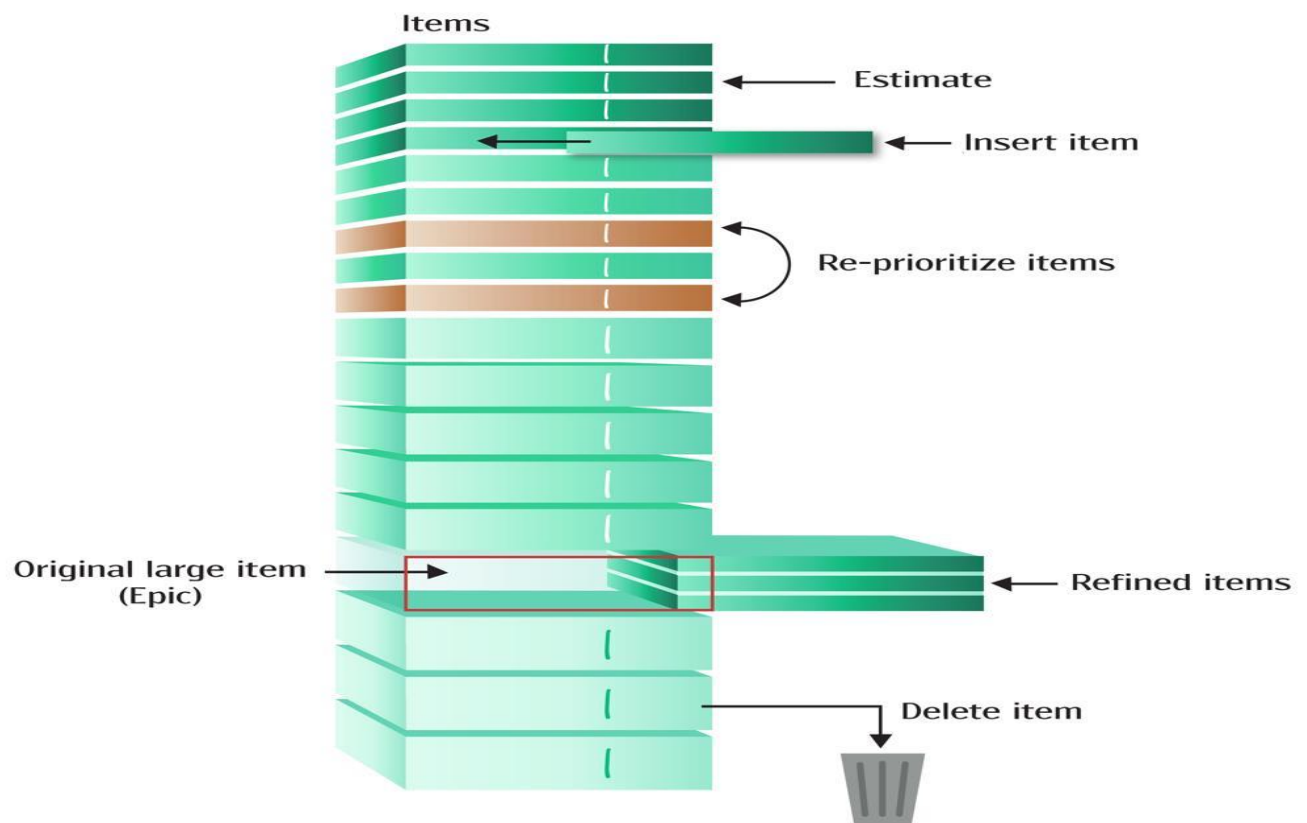
Priority Product Backlog Item
Low Analytics Dashboard

Product Backlog





PRODUCT BACKLOG



5. Sprint Backlog (Sprint 1)

Task	Assigned Team
Design UI	Frontend Team
Customer Registration	Backend Team
Login Authentication	Backend Team
Blockchain Integration	Blockchain Team
Smart Contract Development	Blockchain Team
Database Design	Database Team
API Development	Backend Team
Testing	QA Team

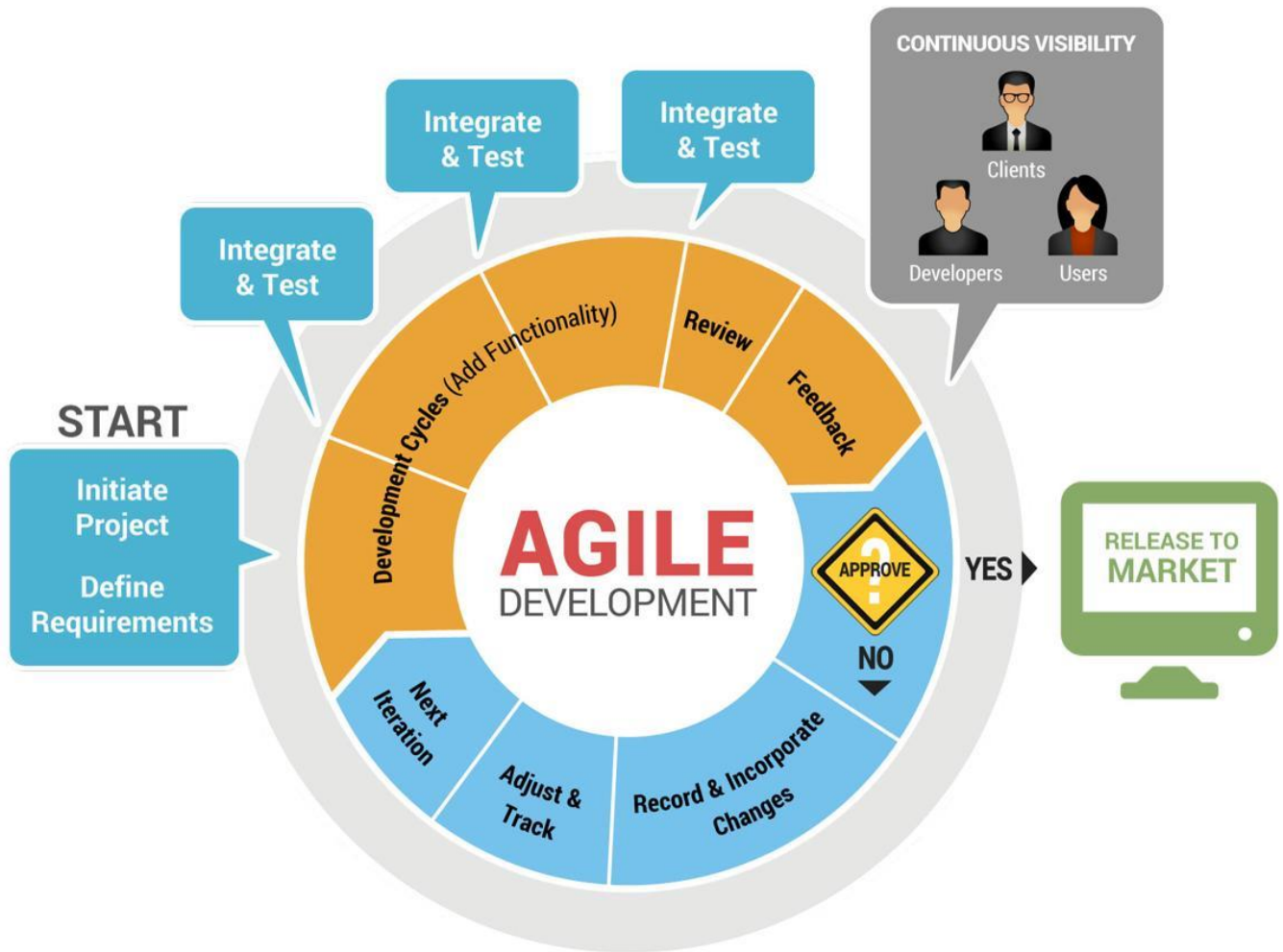
Sprint Duration

2 Weeks (14 Days)

Sprint Tasks

- UI Design
 - Login Module
 - Registration Module
 - Blockchain Setup
 - Smart Contract Creation
 - Transaction Module
 - API Development
 - Unit Testing
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Scrum Sprint Planning



6

6. Story Point Estimation

Using Fibonacci Estimation

User Story	Story Points	Justification
Registration	3	Simple
Login	5	Security Implementation
Fund Transfer	8	Blockchain Validation
Transaction History	3	Medium Complexity
Smart Contracts	13	High Complexity
Fraud Detection	8	AI & Security Logic
Reports	5	Moderate Complexity

Total Story Points

45 Story Points

Agile Story Points





7. Sprint Goal

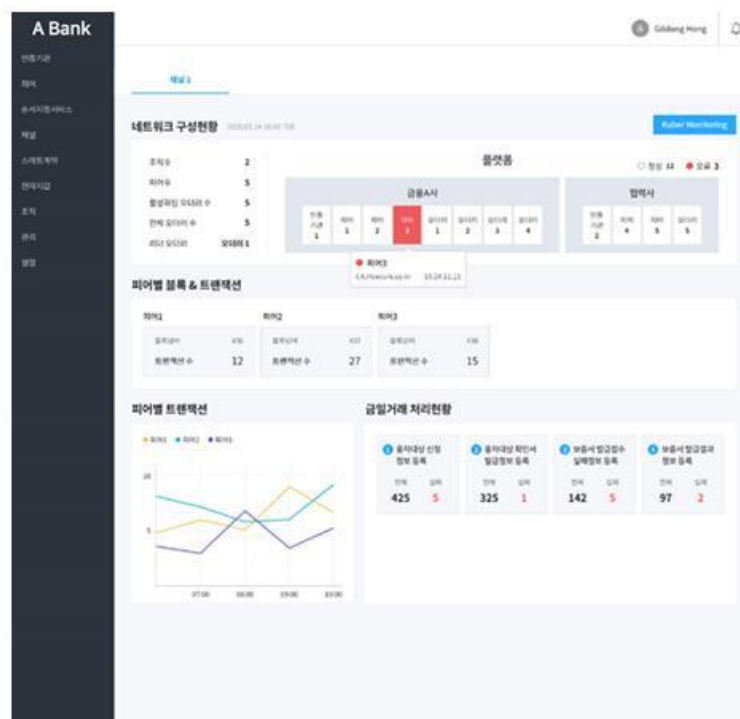
Sprint Goal

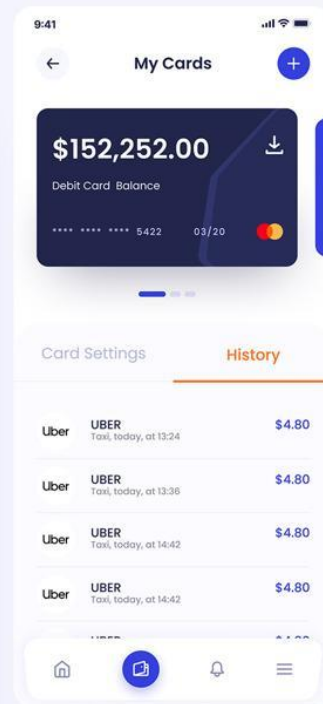
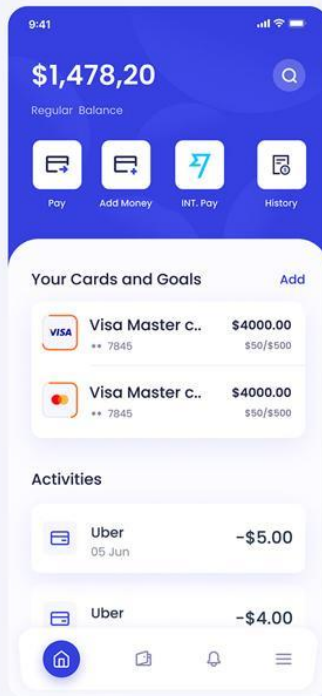
Develop the core modules of the Blockchain-Based Secure Banking Transaction Management System, including customer registration, secure login, blockchain integration, smart contract execution, and secure fund transfer functionality.

Expected Sprint Deliverables

- Customer Registration Module
- Secure Login Module
- Blockchain Network Setup
- Smart Contract Module
- Fund Transfer Module
- Blockchain Transaction Ledger
- Database Design
- REST APIs
- Unit Testing Report
- Sprint Review Document

Final Banking Solution







Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification	/5
Epics & User Stories	/5
Acceptance Criteria	/5
Product Backlog Prioritization & Sprint Planning	/5
Story Point Estimation, Sprint Goal & Presentation	/5
TOTAL	/25